Bachelor of Computer Application
Semester – V

Sub: Business Analytics

Topic: Importing business datasets from flat files & MySQL and connecting with any other sources into Power BI

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Importing Business Datasets from Excel/CSV & MySQL and Connecting with Other Sources into Power BI

Problem Statement

AdventureWorks, a retail company, stores its data across flat files (Excel/CSV) and a MySQL database. As a Data Analyst, the task was to connect Power BI Desktop to these sources, import Sales, Customer, and Product data, and build visuals to analyze sales by product, category, customer, region, and time. The goal was a centralized dashboard for business reporting.

Methodology / Steps Followed

Step 1: Data Collection

- Sales Data (AdventureWorks Sales Data 2022) collected as a CSV file.
- Customer Data (AdventureWorks Customer Lookup) collected as a CSV file.
- Product Data collected from three linked CSV files: Product Lookup, Product Subcategories Lookup, and Product Categories Lookup.
- Territory Data (AdventureWorks Territory Lookup) collected as a CSV file.
- Product Lookup data was also verified against a MySQL database (adventureworks schema) holding the same tables.

Step 2: Data Import

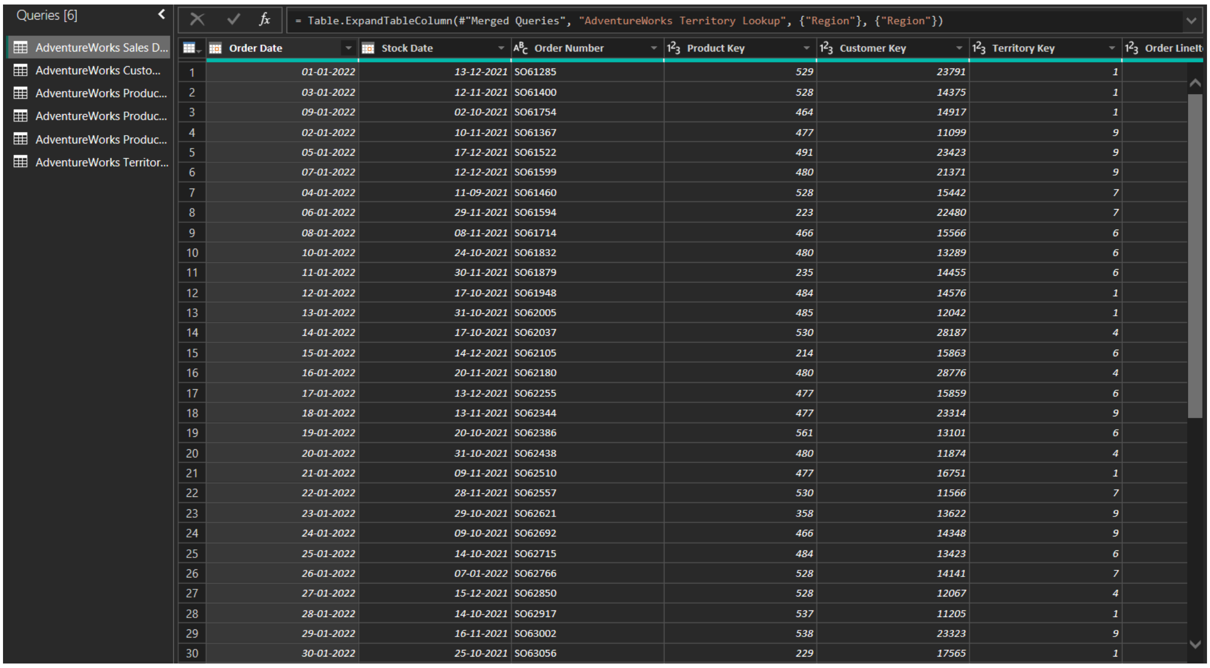
- Opened Power BI Desktop and connected using Get Data → Text/CSV for each flat file.
- Verified the MySQL source using SQL queries (USE adventureworks; SHOW TABLES;).
- Imported all datasets into Power Query Editor via Transform Data.

Step 3: Data Cleaning and Transformation

- Set correct data types for each column (Date, Whole Number, Decimal Number).
- Removed duplicate records from Product Lookup and Customer Lookup.
- Handled missing values in Customer Lookup.
- Merged Product Lookup with Product Subcategories Lookup and Product Categories Lookup to bring in Subcategory Name and Category Name.
- Merged Territory Lookup into Sales Data to bring in the Region field.
- Disabled Enable Load for the Subcategories and Categories staging queries.
- Renamed columns for clarity and consistency.

Fig 1: Territory Lookup merged into Sales Data, adding the Region column

Step 4: Data Modeling



	Order Date	Stock Date	Order Number	Product Key	Customer Key	Territory Key	Order LineIt
1	01-01-2022	13-12-2021	SO61285	529	23791	1	
2	03-01-2022	12-11-2021	SO61400	528	14375	1	
3	09-01-2022	02-10-2021	SO61754	464	14917	1	
4	02-01-2022	10-11-2021	SO61367	477	11099	9	
5	05-01-2022	17-12-2021	SO61522	491	23423	9	
6	07-01-2022	12-12-2021	SO61599	480	21371	9	
7	04-01-2022	11-09-2021	SO61460	528	15442	7	
8	06-01-2022	29-11-2021	SO61594	223	22480	7	
9	08-01-2022	08-11-2021	SO61714	466	15566	6	
10	10-01-2022	24-10-2021	SO61832	480	13289	6	
11	11-01-2022	30-11-2021	SO61879	235	14455	6	
12	12-01-2022	17-10-2021	SO61948	484	14576	1	
13	13-01-2022	31-10-2021	SO62005	485	12042	1	
14	14-01-2022	17-10-2021	SO62037	530	28187	4	
15	15-01-2022	14-12-2021	SO62105	214	15863	6	
16	16-01-2022	20-11-2021	SO62180	480	28776	4	
17	17-01-2022	13-12-2021	SO62255	477	15859	6	
18	18-01-2022	13-11-2021	SO62344	477	23314	9	
19	19-01-2022	20-10-2021	SO62386	561	13101	6	
20	20-01-2022	31-10-2021	SO62438	480	11874	4	
21	21-01-2022	09-11-2021	SO62510	477	16751	1	
22	22-01-2022	28-11-2021	SO62557	530	11566	7	
23	23-01-2022	29-10-2021	SO62621	358	13622	9	
24	24-01-2022	09-10-2021	SO62692	466	14348	9	
25	25-01-2022	14-10-2021	SO62715	484	13423	6	
26	26-01-2022	07-01-2022	SO62766	528	14141	7	
27	27-01-2022	15-12-2021	SO62850	528	12067	4	
28	28-01-2022	14-10-2021	SO62917	537	11205	1	
29	29-01-2022	16-11-2021	SO63002	538	23323	9	
30	30-01-2022	25-10-2021	SO63056	229	17565	1	

- Related Sales Data 2022[Product Key] to Product Lookup[Product Key] (one-to-many).
- Related Sales Data 2022[Customer Key] to Customer Lookup[Customer Key] (one-to-many).
- Verified all relationships and cardinality in Model View.

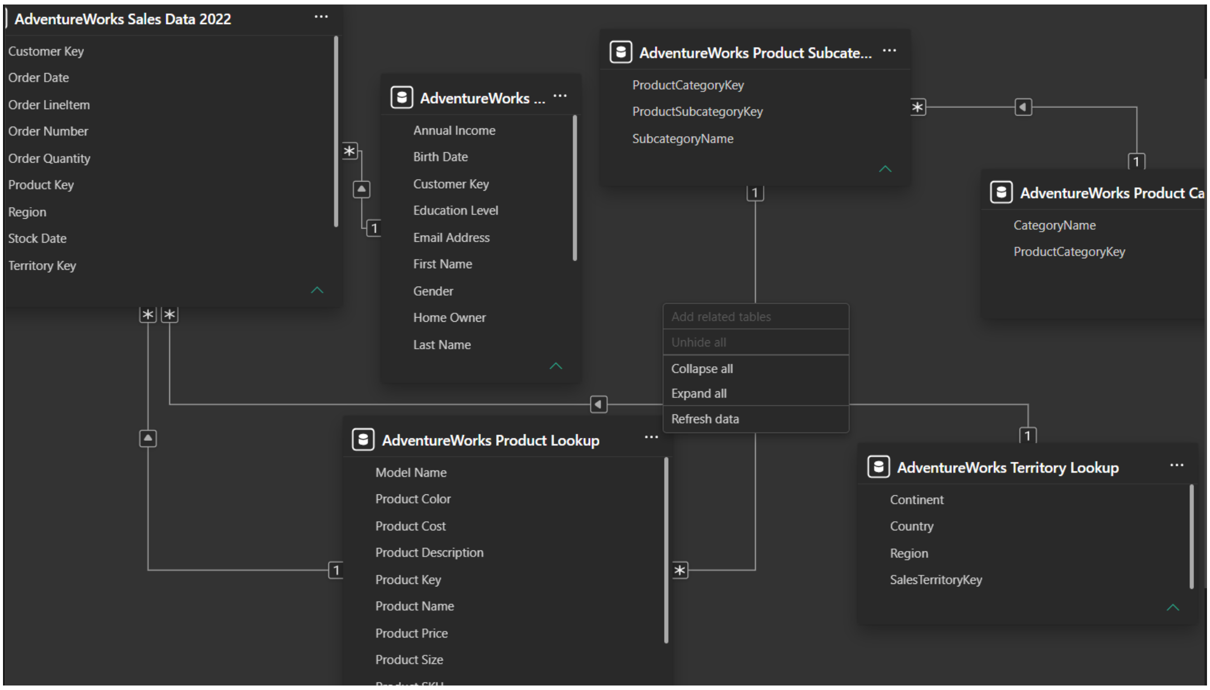


Fig 2: Power BI Model View showing relationships between Sales, Product, Subcategory, Category, and Territory tables

Step 5: Dashboard Development

- Created measures: Total Sales, Total Orders, and Total Customers (from Sales Data).
- Created an extra measure, Total Customers (Table), for the full customer base.

- Built KPI cards for Total Orders, Total Customers, and Total Sales.
- Built a bar chart for Total Sales by Category Name.
- Built a bar chart for Total Sales by Region.
- Built a line chart for Total Sales by Order Date.
- Added an Order Date slicer to filter the dashboard.

Output

The final Power BI dashboard provides:

- Total Sales Performance — KPI cards for Total Orders (181), Total Customers, and Total Sales (22.20K).
- Customer Analysis — Total Customers based on actual sales transactions.
- Product-wise Sales Analysis — Total Sales by Category, led by Bikes, then Accessories and Clothing.
- Regional Sales Performance — Total Sales by Region, led by Australia, Southwest, United Kingdom, Canada, and Northwest.
- Monthly Sales Trends — Total Sales by Order Date, with recurring monthly spikes.
- Interactive Filters and Slicers — a date-range slicer on Order Date.

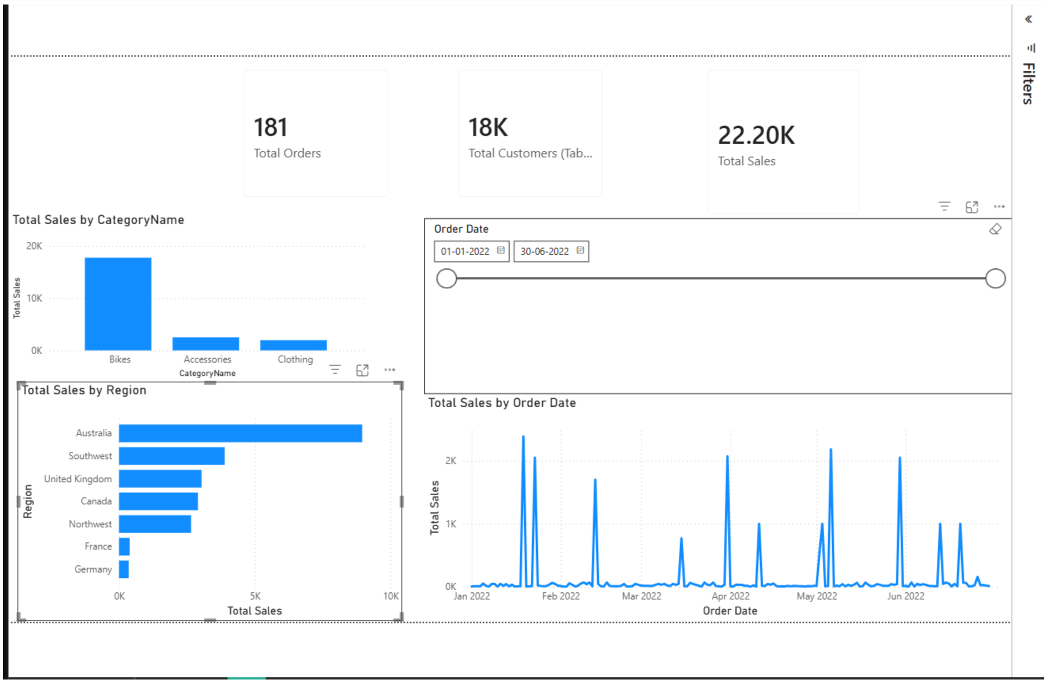


Fig 3: Final AdventureWorks Sales Dashboard in Power BI

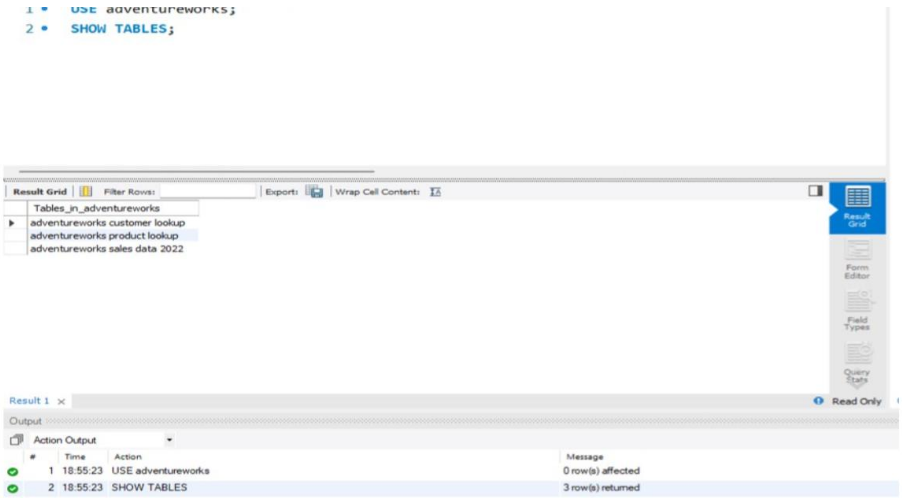


Fig 4: MySQL database (adventureworks schema) verified as an additional data source

The dashboard helps management track business performance and make data-driven decisions.

Conclusion

This project integrated data from CSV files and a MySQL database into Power BI. Cleaning, merging, and modeling steps were applied to build an accurate, centralized dashboard. It gives a clear view of sales across products, regions, customers, and time, supporting better decision-making.

Learning Outcomes

Through this project, I learned:

- How to import data from CSV files and a MySQL database into Power BI.
- Data cleaning and transformation using Power Query.
- Fixing data type mismatches that caused merge errors.
- Using Merge Queries to bring lookup data into fact tables.
- Using Enable Load to manage staging queries.
- Creating relationships and checking cardinality in Model View.
- Writing DAX measures with SUMX, DISTINCTCOUNT, and COUNTROWS.
- Designing dashboards with cards, charts, and slicers.